P.S.R. ENGINEERING COLLEGE, SIVAKASI – 626 140

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DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING
QUESTION BANK

Programme:	B.E.	Branch	Electronics and Communication Engineering
Academic Year:	2025-2026	Year/Semester	II/IV
Course Code:	231EC45	Course Name	JAVA Programming
Course Tutor:	Mr.D.Arun Shunmugam, AP/ECE Ms.A.Madhumathi AP/ECE		

UNIT –II JAVA CLASSES

PART - A			
1.	Which of these keywords are used to define an abstract class? a)Abstract b)abstract c)abst d)static Answer: b	K1	CO2
2.	In JAVA, can a class implement multiple interfaces? Interfaces cannot be implemented by classes Only abstract classes can implement interfaces No, a class can implement only one interface Yes, a class can implement multiple interfaces Answer: d	K1	CO2
3.	Which of these packages contains abstract keyword? a) java.lang b) java.util c) java.io d) java.system Answer: a	K1	CO2
4.	Arrays in Java are implemented as? a) class b) object c) variable d) none of the mentioned Answer: b	K1	CO2
5.	_____ keyword is used to prevent content of a variable from being modified? a) final b) last c) constant d) static Answer: a	K1	CO2
6.	_____ cannot be declared static? a) class b) object c) variable d) method Answer: b	K1	CO2
7.	Which of these access specifiers can be used for a class so that its members can be accessed by a different class in the different package? a) Public b) Protected c) Private d) No Modifier Answer: a	K1	CO2

8.	What will be the output of the following Java program?
	package pkg;
	class output
	{
	public static void main(String args[])
	{
	StringBuffer s1 = new StringBuffer("Hello");
	s1.setCharAt(1, x);
	System.out.println(s1);
	}
	}
	a) xello b) xxxxx c) Hxllo d) Hexlo Answer: c
		K1	CO2

9.	When does Exceptions in Java arises in code sequence? a) Run Time b) Compilation Time c) Can Occur Any Time d) None of the mentioned Answer: a	K1	CO2
10.	Which of these keywords is used to manually throw an exception? a) try b) finally c) thrown d) catch Answer: c	K1	CO2
11.	What is an exception in Java? a) An error that occurs during compilation b) An unexpected event that occurs during program execution c) A logical error in the code d) A warning issued by the compiler Answer: b	K1	CO2
12.	Which class is the superclass of all exceptions in Java? a) Error b) Runtime Exception c) Throwable d) Exception Answer: c	K1	CO2
13.	Which block must follow a try block? a) final b) catch c) done d) except Answer: b	K1	CO2
14.	Which of the following is used to group multiple exceptions in a single catch block? a) Using the | operator b) Using the & operator c) Using the && operator d) Using the || operator Answer: a	K1	CO2
15.	What is the use of try & catch? a) It allows us to manually handle the exception b) It allows to fix errors c) It prevents automatic terminating of the program in cases when an exception occurs d) All of the mentioned Answer: d	K1	CO2
16.	Which of these keywords are used for the block to be examined for exceptions? a) try b) catch c) throw d) check Answer: a	K1	CO2
17.	What will be the output of the following program? int x=5, y=10; System.out.println(x+y);	K1	CO2

	a) 5 b) 10 c) true d) None of the above Answer: d		
18.	Which of the following is valid syntax to declare an array in Java? a) int arr[] b) arr int[] c) array int d) int[] arr; Answer: a	K1	CO2
19.	Which operator is used for concatenation of Strings in Java? a) + b) & c) && d) % Answer: a	K1	CO2
20.	Which data type is used to store a single character in Java? a) char b) String c) byte d) Boolean Answer: a	K1	CO2

PART-B (2 marks)			
1.	What are different types of access modifiers (Access specifiers)?	K1	CO2
2.	How to define and import a package in Java.		
3.	Define Abstract class	K1	CO2
4.	Compare static and inner class.	K2	CO2
5.	Explain the usage of Java packages.	K2	CO2
6.	What is a package? How to define a package?	K1	CO2
7.	What is the difference between this () and super ()?	K2	CO2
8.	List the Wrapper class in Java.	K1	CO2
9.	What is Exception Handling in Java?	K2	CO2
10.	What is the use of try and catch block in java?	K2	CO2
11.	What are runtime exceptions?	K1	CO2
12.	Demonstrate how to declare a Book class with title and author attributes, create an object, and print the values.	K2	CO2
13.	Demonstrate using throw and throws in a simple method.	K2	CO2
14.	Demonstrate how to define and import a package.	K2	CO2
15.	Explain how to create a Java package named library with classes Book and Member	K2	CO2
16.	Develop a Java program that uses an interface to implement multiple behaviours for a Vehicle class.	K3	CO2
17.	Show how to import the Book class from the library package in another class.	K2	CO2
PART-C (14 marks)			
1.	Explain the concept of abstract classes in Java, describe their properties, and write a program where an abstract class Shape is extended by Circle and Rectangle classes. Display the area of each shape.	K2	CO2
2.	Build a Java Program to implement method Overloading and method overriding	K3	CO2
3.	Explain inner classes in Java and their importance. Write a program with a class Student having an inner class Address. Display student details along with the address.	K2	CO2
4.	Explain **interfaces** in Java and how they are implemented and Create an interface Library with methods drawbook (), return Book(), check Status(), and reserve Book(), and implement it in a class City Library.	K2	CO2

5.	Demonstrate the exception handling mechanism in Java using a program with try, catch, and finally blocks to handle Arithmetic Exception and ArrayIndexOutOfBoundsException.	K2	CO2
6.	How to handle multiple catch blocks for a nested try block with a suitable example program.	K1	CO2
7.	Summarize the uses of throw and throws clauses in Java exception handling and give suitable example program.	K2	CO2
8.	Develop a Library interface which has draw book (), return book () (with fine), check status () and reserve book () methods. All the methods tagged with public.	K3	CO2
9.	Develop a static inner class called Pair which has a minMax method for finding min and max values from an array.	K3	CO2
10.	Explain the Java exception hierarchy with a diagram and give examples of checked and unchecked exceptions.	K2	CO2